Identification of an Unknown – B

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November 30, 2023

Chem 2081 007

Introduction

In organic chemistry, an unknown structure can be identified using different types of spectroscopy. Spectroscopy is important because it allows new compounds to be identified or even old compounds' identity to be confirmed. Not only that, but spectroscopy can also be used in a lab to confirm the purity of a product to help get more accurate results in an experiment. In this lab, several different types of spectroscopy were studied including nuclear magnetic resonance, infrared spectroscopy, and mass spectroscopy.

Body

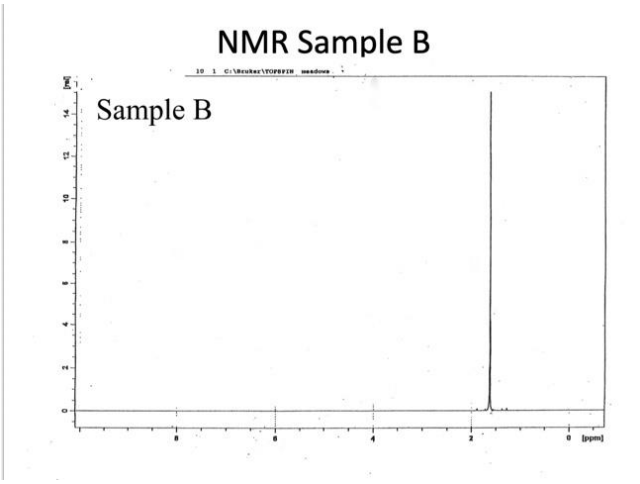
Nuclear magnetic resonance is a method of spectroscopy that involves studying the atomic nucleus and how it interacts with an external magnetic field. This method of spectroscopy is extremely useful in identifying unknown compounds because it provides essential information about the structure. Nuclear magnetic resonance can even be used by itself to determine the structure of an unknown compound without the use of other methods. On the spectrum, signals are recorded when energy is released, which can then be further analyzed. These signals can provide vital information needed to determine an unknown compound's structure. Some key details to consider while interpreting the signals include the location of each signal, the area under each signal, and the shape of the signal. If only one signal is present, this means the compound only contains one kind of proton. The presence of only one signal also shows that the compound will be chemically equivalent. Another essential part of the spectrum includes integration values. The integration values can be used to determine the number of hydrogens at each location. The final characteristic of each signal is its multiplicity, which is defined by how many peaks are in the signal. The multiplicity of each signal indicates the number of hydrogens that are on the neighboring atoms.

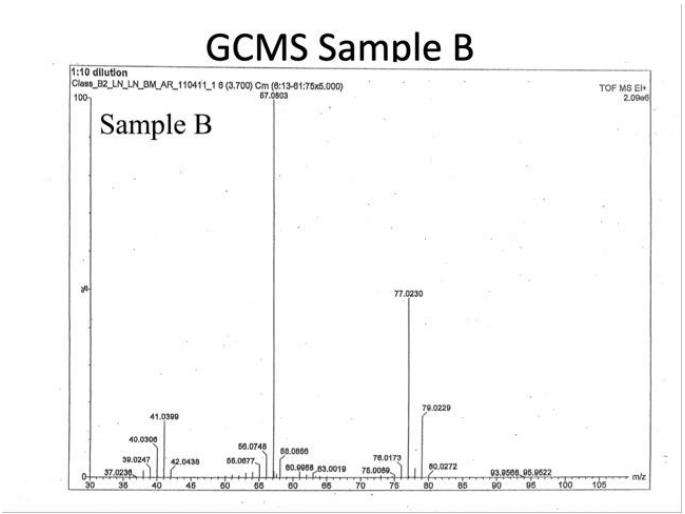
Another method of spectroscopy is infrared spectroscopy. This type of spectroscopy is used to identify which functional groups are present in a compound. Infared spectroscopy works by infrared energy passing through a sample which is then measured as frequency or wavelength. The strength of the absorption is indicated on a graph and is known as an absorption band. An absorption band represents the frequency at which a molecular vibration occurs. These absorption bands can be analyzed to determine what functional groups are present in the compound. To decide which functional groups are in the compound, the region the absorption band is located must be examined. The position of the vibrations can reveal the major functional groups that can be present in a compound. In order to read the infrared spectra correctly, the intensity of the vibrations must be determined. Some of these vibrations are weak, and some of them are very strong. The intensity and shape of the vibrations also helps determine which functional groups are present. According to page 155 of the CHEM 2081 Organic Chemistry II Lab Manuel (Keller, 2023), a procedure can be followed to successfully interpret the infrared spectra. To achieve this, different regions on the infrared spectra are analyzed to determine which functional groups are present, and which functional groups are not.

The final method of spectroscopy being studied is mass spectroscopy. In mass spectroscopy, the interaction between an energy source and matter is being studied. This type of spectroscopy involves a compound being turned into ions, which are deflected depending on the masses of the particles. Mass spectroscopy is used to determine the molecular formula and molecular weight of a compound. In order to do this, the spectra must be read correctly. The spectra must be read by analyzing the peaks and the fragments. To start reading the spectra, the base peak must be determined. The base peak, which is the tallest peak on the spectra, represents the bulkiest part of the molecule. To decide where to stop reading on the spectra, the last largest peak while reading to the right side must be determined. Once the largest peaks are found, they

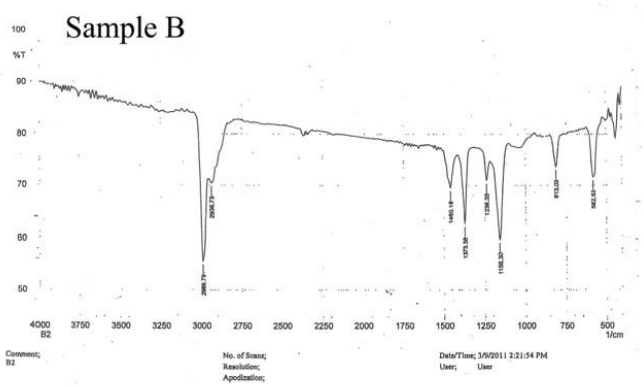
can be used to determine which group in the molecule is the most abundant. To find out what the molecular formula is, the “Rule of Thirteen” can be used. The “Rule of Thirteen” is described on page 161 of the CHEM 2081 Organic Chemistry II Lab Manuel (Keller, 2023). This method uses the m/z of the molecular ion to create the chemical formula of a hydrocarbon. This method of spectroscopy might not provide as many details on the structure of the specific compound, but it can be used with nuclear magnetic resonance and infrared spectroscopy to provide additional information about the compound.

Data





FTIR Sample B



Analysis

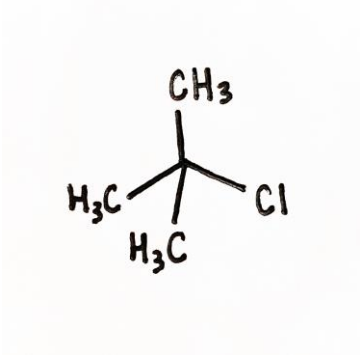
There are several observations that can be made by looking at the NMR. Based on the NMR, there is only one intense peak, which is roughly located around 1.5 ppm. Since there is only one peak, all of the hydrogens in this compound must provide the same signal, so the protons are of the same kind. Therefore, this compound is chemically equivalent.

By looking at the FTIR and using the procedure provided in the appendix of the CHEM 2081 Organic Chemistry II Lab Manuel (Keller, 2023), it is indicated that there is a C-Cl, C-H, and CH_3 bond within the molecule. From the FTIR, it is known that there is a C-Cl stretch due to a peak being present at 582.03 cm^{-1}. Additionally, it is known that there is not a hydroxyl group, because there is not a strong peak in the 1300-1000 cm^{-1} region. This compound also contains C-H bond stretching since there is a strong peak near 3300 cm^{-1}. Using the information obtained from the NMR, in addition to the C-Cl, C-H, and CH_3 bonds, it is known that the compound must have nine hydrogens and symmetry.

Based on the mass spectroscopy graph, the chemical formula of the compound can be found. According to the graph, the base peak is at 57.0803 m/z. The base peak is where the graph should start being read. In this mass spectra, the base peak represents the formula C_4H_9, which is the bulkiest part of the molecule. This was determined by using the "Rule of Thirteen", found on page 155 of the CHEM 2081 Organic Chemistry II Lab Manuel (Keller, 2023). The last largest peak to the right, which is at 79.0229 m/z, represents where to stop reading the mass spectra. In this mass spectra, the last largest peak represents the chlorine in the compound. The peak at 77.0230 m/z represents the CH_3 bonds in the compound. The peak at 77.0230 m/z is higher than the peak at 79.0229 m/z, meaning that the CH_3 group is more abundant in the compound.

Conclusion

Nuclear magnetic resonance, infrared spectroscopy, and mass spectroscopy are all different ways that an unknown compound can be identified. All of these methods can be used on their own to help identify a structure, but they are most helpful when they are used together. Each of these methods of spectroscopy provides different information about the structure, but they can all help identify the unknown sample. After analyzing the NMR, FTIR, and mass spectra, it is known that the unknown sample is tert-butyl chloride, or C_4H_9Cl. This structure has the IUPAC name of 2-chloro-2-methylpropane.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.